Statistical Analysis and Machine Learning-Based

Prediction of Wine Quality

Physicochemical properties • statistics • hypothesis testing • sampling • machine learning

ORIGINAL DATA

1,599

observations

CLEAN DATA

1,359

after 240 duplicates

PREDICTORS

11

physicochemical variables

BEST MODEL

Random Forest

RMSE 0.6440

Primary dataset: UCI Red Wine Quality — dataset page

1. Dataset and Preprocessing

What we started with and what we changed

ROWS	COLUMNS	MISSING VALUES	DUPLICATES
1,599	12	0	240
original	11 features + target	no imputation	exact duplicate rows

Dataset definition

- 11 physicochemical variables + quality score.
- Quality is numerical, so the main ML task is regression.
- Quality scores are concentrated around 5 and 6; scores 3 and 8 are rare.

Cleaning decision

- No missing values → no imputation.
- 240 exact duplicate rows removed to avoid repeated identical records receiving extra weight.
- Final analytical dataset = 1,359 observations.
- Duplicates were not automatically assumed to be errors.

Primary dataset: UCI Red Wine Quality — dataset page

2. Statistics and Hypothesis Testing

What we observed and what the tests changed

MEAN ALCOHOL

10.4323%

cleaned data

95% CI

10.3747–10.4899

95% confidence

ONE-SAMPLE TEST

t = 14.7284

$p = 1.16 \times 10^{-45}$

CORRELATION

r = 0.4803

alcohol vs quality

Test	H_0	Result	Decision
One Sample Two tailed t-test	μ = 10%	$p = 1.16 \times 10^{-45}$	Reject H_0
Two Sample two tailed t-test	μlow = μhigh	$p = 1.80 \times 10^{-69}$	Reject H_0

Low vs high quality

- Low (≤5): mean alcohol = 9.9208%
- High (≥6): mean alcohol = 10.8876%
- Difference = 0.9669 percentage points

What did we learn from increasing sample size?

- Larger samples give more stable estimates.
- Confidence intervals become narrower.
- It becomes easier to detect a difference from 10%.

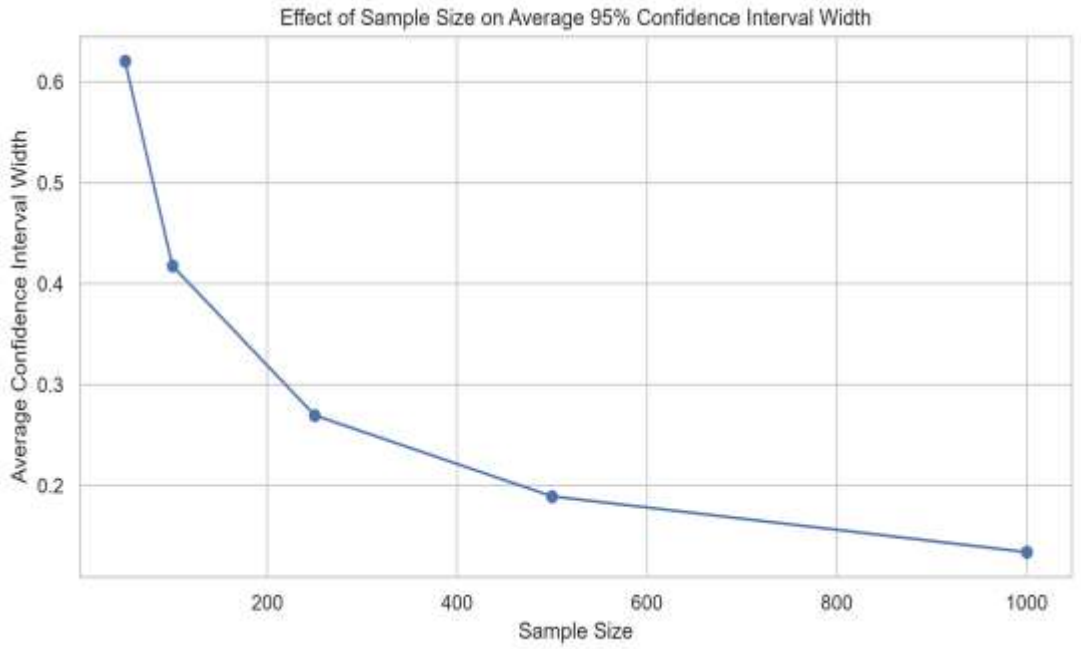
3. Sampling: Population Signal vs Random-Sample Variation

Repeated sampling from the cleaned dataset

n	Mean of sample means	SD of sample means	Avg. CI width
50	10.4453	0.1626	0.6205
100	10.4221	0.1137	0.4177
250	10.4404	0.0691	0.2695
500	10.4201	0.0445	0.1893
1000	10.4325	0.0160	0.1340

What can we conclude?

- Sample means vary because random samples are different.
- Variation shrinks strongly as n increases.
- The sample means center near the cleaned-data mean of 10.4323%.
- Therefore, sampling randomness explains differences among individual samples.



4. Machine Learning Performance

Predicting quality with 5-fold cross-validation

Model	MAE ↓	MSE ↓	RMSE ↓	R^2 ↑
Linear Regression	0.5118	0.4417	0.6645	0.3465
Random Forest	0.4931	0.4150	0.6440	0.3853

Best model

Random Forest

RMSE = 0.6440
R^2 = 0.3853

Random Forest feature importance

Feature	Importance
Alcohol	0.2777
Sulphates	0.1392
Volatile acidity	0.1295
Total sulfur dioxide	0.0770
Chlorides	0.0623

Observations

- Random Forest outperformed Linear Regression on all four metrics.
- It can capture nonlinear relationships and interactions.
- Alcohol is the strongest model feature.
- R^2 = 0.3853 is not 38.5% prediction accuracy.

5. Conclusions, Limitations and Recent Research

What the project ultimately tells us

Main conclusions

- Alcohol is positively associated with quality and is the most important Random Forest feature.
- Both hypothesis tests reject H_0 at $\alpha = 0.05$.
- Larger samples give more stable means and narrower CIs.
- Random Forest performs better than Linear Regression.

Limitations

- Observational data → no causal claim.
- The UCI dataset is not automatically the entire real-world population.
- $R^2 = 0.3853$ leaves substantial variation unexplained.

Recent study — 2025

Khan et al. (2025/2026), "Improved machine learning framework with feature engineering and SHAP analysis for predicting wine quality"

Discover Applied Sciences. Published online 29 December 2025. The study uses the UCI wine-quality data and identifies alcohol, sulfur dioxide and volatile acidity among influential predictors.

Why it matters here

Our project reaches a compatible feature-level observation: alcohol is strongly related to quality and is the top Random Forest feature.

Primary dataset: UCI Wine Quality - dataset page

Project Repository – Github link

Cortez et al. (2009) — original UCI wine-quality study.